# MILITARY SPECIFICATION

# PAPER, PRINTING, PLASTIC COATED

## 1. SCOPE

1.1 Scope.-This specification covers one type of plastic fiber base, plastic mineral coated printing paper (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-31 - Paper, General Specifications and Methods of Testing
CCC-T-191- Textile Test Methods

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

STANDARDS

FEDERAL

Fed. Test Metnod Std. No. 101 - Preservation, Packaging, and Packing Materials: Test Procedures

FSC: 8140

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D-585-68 - Sampling and Accepting a Single Lot of Paper, Paperboard, Fiberboard or Related Product
D641-49 - Conditioning Paperboard, Fiberboard and Paperboard Containers for Testing
D-645-67 - Test for Thickness of Paper and Paperboard
D-646-67 - Basis Weight of Paper and Paperboard
D-685-44 - Conditioning Paper and Paper Products for Testing
D-918-49 - Blocking Resistance of Paper and Paperboard

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Materials.-The material shall consist of a suitably compounded plastic fiber base sheet coated on both sides with a suitably compounded plastic mineral coating. The fiber portion of the base sheet shall be free from unbleached and ground wood pulp.

3.2 Printing characteristics.-The paper shall be suitable for multicolor face and back letter press and offset printing. The paper shall be suitable for use on web fed presses and shall be clean cutting and static free after printing.

3.3 Form and dimension.-The material shall be furnished in flat cut sheets or in rolls (see 6.1). Flat cut sheets shall be of the length and width indicated, the tolerance being not more than 1/8 inch oversize unless otherwise specified. Undersized paper will not be considered acceptable. Additionally, the paper must be flat and cut square on all four sides. Rolls shall be of the desired width and length (net weight or outside diameter) wound on cores of the specified construction



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and inside diameter. Unless otherwise indicated, the width must not be less than specified and not more than 1/8 inch oversize. When the material is ordered to narrow width, a sufficient number of strips shall be wound on a common core to produce a roll having a convenient combined width, as specified.

3.4 Stability

3.4.1 General.-The material shall not discolor, become brittle or be subject to attack by mold, mildew or fungus under normal storage conditions. Paper shall be uniform, free from mottle and any particles which will pick, lift, fluff or pile on the blanket under normal press conditions.

3.4.2 Coating.-The base sheet shall be coated on both sides so as to produce a product which shall not flake, powder, blister or otherwise deteriorate under printing or other normal handling conditions.

3.5 Low temperature characteristics.-The paper shall not become brittle and shall not break when folded once through 180 degrees at minus (-) 40 degrees (°) Fahrenheit (F). A slight stiffening of the paper and coating cracks within 1/2 inch(") of the line of the 180° fold shall not be considered cause for rejection.

3.6 Color.-The color shall be white unless otherwise specified.

3.7 Curls.-The paper must lie flat with no tendency to curl.

3.8 Splices.-Paper supplied in roll form shall have a maximum of 3 splices per roll and each splice shall be conspicuously marked with a red marker at the splice.

3.9 Appearance.-The paper shall have a super-calender finish on both sides and shall conform to the standard samples accompanying the invitations for bids with respect to finish, formation and cleanliness.

3.10 Grain.-Direction of grain on flat paper shall be as ordered.

3.11 Physical properties.-The paper shall comply with the requirements of Table I when tested as specified in Section 4.



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Table I

Physical and Chemical Requirements

| | |
|---|---|
| Basis Weight, lbs per 25 x 38 (1,000 sheets) (tolerance plus or minus (±) 10 percent) | 160 lbs. |
| Thickness, inches (tolerance plus or minus 5 percent) | .0042 in. |
| Tearing resistance-Average each direction min. | 50 grams |
| Brightness, percent minimum | 74 percent |
| Opacity - Average, not less than | 95 percent |
| Tensile breaking strength, dry, each direction min. per inch width | 11 lbs. |
| Elongation at rupture, dry, percent min | |
| Machine direction | 5.5 percent |
| Cross direction | 9.0 percent |
| Tensile breaking strength, wet, min each direction shall be | 4 lbs/inch width |
| Blocking resistance, 140°F and 77 percent relative humidity | shall not be more than slight (see 6.7) |
| Folding endurance M.I.T. double folds, each direction min. | 120 |
| Reflectance drop percent max. | 4.0 |
| Acidity, pH cold, minimum | 7.0 |

3.12 Workmanship.-The paper shall be manufactured in accordance with first grade practice to produce material of uniform quality substantially free from defects of any character.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.



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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the age t's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates, unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and snall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided



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the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.2.2 Lot size.-A lot shall consist of 5 rolls or 5 units of sheets or fraction thereof, which has been manufactured from the same types and proportions of ingredients by the same process by one manufacturer, and submitted for acceptance at one time.

4.2.3 Examination.-Random samples shall be drawn from each lot for examination of material, workmanship, dimensions, count, and preparation for delivery defects, using the inspection levels and acceptable quality levels (AQL's) specified in 4.3.1. The lot size for purpose of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of rolls or packages of sheets, as applicable, and in units of shipping containers for examination under 4.3.1. For this purpose, two or more strips slit from the same original sheet and rewound side by side onto another common core shall be considered one roll, except that for examination for slit width, each strip shall be considered a separate "narrow" roll.

4.3 Testing

4.3.1 Sampling.-Unless otherwise specified, the sampling plan for each of the following inspections and tests shall be in accordance with MIL-STD-105, Table IIA, using Code letter D and an AQL of 0.40 for each major defect and an AQL of 0.65 for each minor defect .

| Categories | Defects | Method Inspection | Code No. (see 6.8) |
|---|---|---|---|
| Fiber Identification and Quantitative Determination | Major | Test | 01001 |
| Tearing Resistance | Major | Test | 01002 |
| Tensile Strength (Dry) and Elongation | Major | Test | 01003 |
| Tensile Strength (Wet) | Major | Test | 01004 |
| Folding Endurance, MIT | Major | Test | 01005 |



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| | | | |
|---|---|---|---|
| Acidity pH | Major | Test | 01006 |
| Machine Direction (Grain) | Major | Test | 01007 |
| Basis Weight | Major | Balance(CD) | 01008 |
| Opacity | Minor | Test | 01009 |
| Brightness | Minor | Test | 01010 |
| Thickness | Minor | Gage(CD) | 01011 |
| Form and Dimension | Minor | Gage(CD) | 01012 |
| Color | Minor | Visual | 01013 |
| Workmanship | Minor | Visual | 01014 |

4.3.2 Blocking resistance, Major defect, Code No. 02001.-Test specimens shall be selected in accordance with the provisions of ASTM-D-918-49. Failure to meet the applicable requirement when tested as specified in 4.4.1 shall be cause for rejection of the lot.

4.3.3 Low temperature flexibility - Minor defect, Code No. 03001.-Three samples measuring 3 inches by approximately 8 inches, shall be selected per each lot and tested in accordance with paragraph 4.4.2. Upon completion of testing examine the folded area for rupture of the sample along the line of the fold. Visible breaks extending from one face to the opposite face at any place in the sample or coating failure at any place in the sample or coating failure at any point more than 1/2 inch from the line of fold shall be considered grounds for rejection. Failure of the coating within 1/2 inch of the line of fold shall not be considered grounds for rejection.

4.3.4 Reflectance drop, Minor defect, 04001.-A minimum of one sample pad per lot shall be subjected to the test specified in 4.4.3. A drop of more than 4 percent shall be considered grounds for rejection.

4.3.5 Inspection.-The finished material shall be visually or dimensionally inspected, as applicable, for compliance with the dimensional requirements, and for workmanship (see 4.3.1).

4.3.5.1 Preparation for delivery.-An examination shall be made to determine that marking, materials and workmanship comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior the closing operation. Closed shipping containers shall be examined for closure defects.



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| Examine | Defect |
|---|---|
| Markings (exterior and interior) | Incomplete; incorrect, illegible; omitted; of improper size, location, sequence or method of application (see 5.2) |
| Materials | Any nonconforming component; component missing, damaged or otherwise defective |
| Workmanship | Inadequate application of components such as: incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling<br>Bulged or distorted container |

4.3.6 Test samples.-Samples shall consist of the quantities and size, prepared as specified, in the applicable Standard, ASTM or as indicated in Federal Specification UU-P-31 (Method 160). Each sample unit shall consist of a number of full size sheets or of a length of roll stock full width of the roll sufficient to provide material for making all the tests specified. Unless otherwise specified, no more than one test unit shall be selected form any one roll or package of sheets. From each test unit one set of specimens shall be prepared. This set shall be representative of the full width of the sample unit as obtained either by selecting a set of specimens from locations evenly or randomly spaced along the width of the unit or by compositing material taken from those locations. The specimens shall be conditioned in accordance with ASTM (Designations D641 and D685) and shall be tested as specified. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result (see 4.3.1).

4.3.7 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.3.1 (see 6.3).

4.4 Test Methods and Procedures



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4.4.1 Performance of tests.-The following tests shall be performed in accordance with the pertinent document and method specified:

| Test | Document | Method |
|---|---|---|
| | Fed. Spec. | |
| Machine Direction(Grain) Fiber Identification and Quantitative Determination | UU-P-31 | 140 |
| Brightness | UU-P-31 | 300 |
| Opacity | UU-P-31 | 111 |
| Tearing Resistance | UU-P-31 | 151 |
| Tensile Breaking Strength (Dry) and Elongation | UU-P-31 | 170 |
| Tensile Breaking Strength (Wet) 1/ | UU-P-31 | 171 |
| Folding Endurance,M.I.T. | UU-P-31 | 172 |
| Blocking Resistance | ASTM | 132 |
| Basis Weight | ASTM | D918-49 |
| Thickness | ASTM | D646-67 |
| Acidity,pH | Fed Test Method Std 101 | D645-67 |
| | | 6002 |

1/Apply test to samples that have been heat treated for 4.0 hours in a circulating air oven at 105 degees C. and soaked not less than 72 hours in distilled water at room temperature.

4.4.2 Low temperature flexibility.-A sample 3 in. by approximately 8 in, with the longest edge parallel to the cross direction of the paper shall be conditioned for at least 1 hour at -40 degress F. After conditioning and while still at -40 degrees F. the two 3 in. edges of the sample shall be placed together on a flat horizontal surface and the center of the sample parallel to the 3 in edges shall be folded once through 180 degrees by flattening the sample with a roller. The roller shall be 3 1/2 plus or minus 1/4 inches long, shall weigh 3 plus or minus 0.1 lbs and shall have a suitable handle. The sample, horizontal surface, roller, and the atmosphere immediately surrounding the sample shall be at -40 degrees F. when the sample is flattened.

4.4.3 Reflectance drop



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4.4.3.1 A sample pad of paper, approximately 2 1/2" by 3 in. shall be prepared and the brightness of the paper shall be determined in accordance with UU-P-31, Method 111, Brightness. The portions of the paper used in each individual brightness determination shall be suitably marked so that the brightness of exactly the same portions of the pad can be determined after exposure.

4.4.3.2 The individual sheets of paper from the sample pad shall be placed in suitable sample holders and all the sheets shall be exposed to a carbon arc light for 48 Standard Fading Hours in accordance with CCC-T-191 Method 5600, Colorfastness to Light of Cloth, Accelerated Method (Fade-Ometer). Each sheet from the sample pad shall be exposed separately with the surface used in the brightness determination facing the arc.

4.4.3.3 After the exposure, the sheets shall be reassembled into a sample pad and the brightness of the previously marked portions of the pad shall be determined.

4.4.3.4 The change in the average percent brightness determined before and after exposure shall be reported.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1 Packing.-Unless otherwise specified, packing shall be Level C.

5.1.1 Level C.-Paper shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking for shipment.-Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

PAPER, PRINTING, PLASTIC COATED
FORM_____COLOR_____WIDTH_____LENGTH_____
SPECIFICATION MIL-P-48224 (MU)



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6. NOTES

6.1 Intended use.-The paper covered by this specification is intended for use as a printing material for application, that requires a durable, weather-resistant material.

6.2 Ordering data.-Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Nominal thickness required
c. Length and width of cut sheets
d. Slit width of roll stock
e. No of slit widths to be wound on common core
f. Length, (or net weight or outside roll diameter required).
g. Roll core construction and dimensions required
h. Special dimensional tolerance, if required
i. Preparation for delivery, if required (see Sections 5 and 6)

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment List referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings/which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.



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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3. 1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SMUPA-QA-A-P, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Preparation for delivery.-Rolls or stocks of cut sheets are often shipped without packaging, being protected from the weather by being shipped in box cars or closed vans. When specified, the material is palletized, a pallet load usually weigning approximately one ton.



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6.5 Available sizes.-Paper in roll form is available in widths to 60 in. Sheets are available in dimensions to 60 x 78 inches.

6.6 Standard samples.-Standard samples where needed may be obtained from the procuring agency.

6.7 Definition of Blocking Terms

| Degree of Blocking | Description of Blocking |
|---|---|
| No Blocking -- -- -- | No adhesion or cohesion between contiguous surfaces which slide freely upon one another. Surfaces of specimens are not marred when sheets are separated. |
| Slight Blocking -- -- | Slight adhesion or cohesion between contiguous surfaces. Adjacent surfaces do not slide freely and upon separation of the sheets the surfaces show an occasional evidence of slight marring. |
| Considerable Blocking-- | Adhesion or cohesion of contiguous surfaces. Layers may be separated with difficulty. Surfaces will be definitely marred or partially destroyed upon separation. |

6.8 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collechion and analysis by the Government.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 8140-A109



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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Picatinny Arsenal
Dover, New Jersey 07801

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DEPARTMENT OF THE ARMY
DOD 314

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Commander
Picatinny Arsenal
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